



UNIVERSITY OF NIAGARA FALLS

Master of Data Analytics

Assignment 2: Prescriptive Statistical Problems

Course: Spring 2025 Prescriptive Analytics

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Business Challenge: Cooling Fan Replacement Policy Evaluation in a Data Center Using Simulation

A data center relies on a critical server rack that is cooled by three identical cooling fans. The lifetimes of the fans follow the same probability distribution, as specified in Table 1. When a fan fails, the server shuts down, triggering the dispatch of a repair technician. The technician's arrival delay is a random variable with the probability distribution provided in Table 2. The cost and time associated with fan repairs are as follows

- The cost of each fan is \$32
- The server downtime cost is \$10 per minute.
- The technician's labour cost is \$30 per hour.
- Replacement times are:
 - 1 fan: 20 minutes
 - 2 fans: 30 minutes
 - 3 fans: 40 minutes

A proposed policy suggests replacing all three fans whenever any single fan fails, in order to reduce the frequency of future breakdowns and interruptions.

Management would like to evaluate both policies using a Monte Carlo simulation to estimate the cost implications of servicing 45 fan failures.

Tasks

A. Simulation Model Design in Excel

Design two separate Monte Carlo simulation models in Excel:

1. Current Policy: Replace only the failed fan.

Figure 1

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Trial #	Failure Time	Fan A Life (hrs)	Fan B Life (hrs)	Fan C Life (hrs)	Fans Replaced	Arrival (min)	Repair Time (min)	Downtime (min)	Fan Cost	Tech Cost	Downtime Cost	Total Cost			
2	1	1100	1100	1900	1300	1	30	20	50	32	25	500	557	CURRENT POLICY		
3	2	1300	1200	800	200	1	20	20	40	32	20	400	452			
4	3	1900	1000	600	1400	1	20	20	40	32	20	400	452			
5	4	2300	400	1500	800	1	20	20	40	32	20	400	452			
6	5	2700	1300	1100	400	1	45	20	65	32	32.5	650	714.5			
7	6	3400	900	700	1200	1	20	20	40	32	20	400	452			
8	7	3600	200	1900	500	1	20	20	40	32	20	400	452			
9	8	3900	1300	1700	300	1	20	20	40	32	20	400	452			
10	9	4900	1000	1400	1200	1	20	20	40	32	20	400	452			

- | | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
|----|---------|--------------|------------------|------------------|------------------|---------------|---------------|-------------------|----------------|----------|-----------|---------------|------------|---|---|---|
| 1 | Trial # | Failure Time | Fan A Life (hrs) | Fan B Life (hrs) | Fan C Life (hrs) | Fans Replaced | Arrival (min) | Repair Time (min) | Downtime (min) | Fan Cost | Tech Cost | Downtime Cost | Total Cost | | | |
| 2 | 1 | 1100 | 1300 | 1700 | 1100 | 3 | 30 | 40 | 70 | 96 | 35 | 700 | 831 | | | |
| 3 | 2 | 2300 | 1400 | 1200 | 1400 | 3 | 20 | 40 | 60 | 96 | 30 | 600 | 726 | | | |
| 4 | 3 | 3400 | 1100 | 1400 | 1200 | 3 | 20 | 40 | 60 | 96 | 30 | 600 | 726 | | | |
| 5 | 4 | 4600 | 1300 | 1800 | 1200 | 3 | 30 | 40 | 70 | 96 | 35 | 700 | 831 | | | |
| 6 | 5 | 5600 | 1400 | 1000 | 1400 | 3 | 20 | 40 | 60 | 96 | 30 | 600 | 726 | | | |
| 7 | 6 | 6800 | 1400 | 1200 | 1300 | 3 | 45 | 40 | 85 | 96 | 42.5 | 850 | 988.5 | | | |
| 8 | 7 | 7900 | 1100 | 1300 | 1900 | 3 | 30 | 40 | 70 | 96 | 35 | 700 | 831 | | | |
| 9 | 8 | 8900 | 1200 | 1600 | 1000 | 3 | 20 | 40 | 60 | 96 | 30 | 600 | 726 | | | |
| 10 | 9 | 9900 | 1500 | 1500 | 1000 | 3 | 30 | 40 | 70 | 96 | 35 | 700 | 831 | | | |

- | | A | B | C | D | E | F | G | H | I | J | K | L | M |
|----|---------|--------------|------------------|------------------|------------------|---------------|---------------|-------------------|----------------|----------|-----------|--------------------|-------------|
| 1 | Trial # | Failure Time | Fan A Life (hrs) | Fan B Life (hrs) | Fan C Life (hrs) | Fans Replaced | Arrival (min) | Repair Time (min) | Downtime (min) | Fan Cost | Tech Cost | Downtime Cost | Total Cost |
| 41 | 40 | 18700 | 1800 | 500 | 200 | 1 | 20 | 20 | 40 | 32 | 20 | 400 | 452 |
| 42 | 41 | 19000 | 1600 | 300 | 1500 | 1 | 20 | 20 | 40 | 32 | 20 | 400 | 452 |
| 43 | 42 | 20200 | 1300 | 1200 | 1200 | 2 | 20 | 30 | 50 | 64 | 25 | 500 | 589 |
| 44 | 43 | 20300 | 100 | 1500 | 1200 | 1 | 30 | 20 | 50 | 32 | 25 | 500 | 557 |
| 45 | 44 | 21400 | 1300 | 1400 | 1100 | 1 | 30 | 20 | 50 | 32 | 25 | 500 | 557 |
| 46 | 45 | 21600 | 200 | 300 | 1300 | 1 | 30 | 20 | 50 | 32 | 25 | 500 | 557 |
| 47 | | | | | | | | | | | | | |
| 48 | | | | | | | | | | | | Average Total Cost | 523.8777778 |

- [illegible]

Probability distribution of the operational life of cooling fans.

Life (hrs)	Probability	Cumulative
1000	0.1	0.1
1100	0.13	0.23
1200	0.25	0.48
1300	0.13	0.61
1400	0.09	0.7
1500	0.12	0.82
1600	0.02	0.84
1700	0.06	0.9
1800	0.05	0.95
1900	0.05	1

Table 2

Probability distribution of the technician arrival time.

Arrival (min)	Probability	Cumulative
20	0.6	0.6
30	0.3	0.9
45	0.1	1

B. Python Script for Monte Carlo Simulation

Current Policy: Replace only the failed fan.

```
# --- Current Policy Simulation ---

# Start with 3 fans, each randomly assigned a lifetime
fan_lives = np.random.choice(lifetimes, size=3, p=lifetime_probs)
current_time = 0
results = []

# Run the simulation for a number of cycles
for cycle in range(1, num_cycles + 1):
    time_to_failure = min(fan_lives)
    current_time += time_to_failure

    failed_fans = sum(1 for life in fan_lives if life ==
time_to_failure)    # Count how many fans failed at the same time.
```

```

    # Replace only the failed fans, the others fans continue with reduced
    life
    new_lives = []
    for life in fan_lives:
        if life == time_to_failure:
            new_life = np.random.choice(lifetimes, p=lifetime_probs)
            new_lives.append(new_life)
        else:
            new_lives.append(life - time_to_failure)
    fan_lives = np.array(new_lives)

    # Random arrival time and get repair time depending on how many fans
    failed
    arrival_time = np.random.choice(arrival_times, p=arrival_probs)
    repair_time = repair_times[failed_fans]
    downtime = arrival_time + repair_time

    # Cost
    fan_cost = failed_fans * fan_cost_per_unit
    tech_cost = (downtime / 60) * tech_cost_per_hour
    downtime_cost = downtime * downtime_cost_per_min
    total_cost = fan_cost + tech_cost + downtime_cost

    results.append({
        'Cycle': cycle,
        'Policy': 'Current',
        'Time of Failure': current_time,
        'Fans Replaced': failed_fans,
        'Arrival (min)': arrival_time,
        'Repair Time (min)': repair_time,
        'Downtime (min)': downtime,
        'Fan Cost ($)': fan_cost,
        'Tech Cost ($)': tech_cost,
        'Downtime Cost ($)': downtime_cost,
        'Total Cost ($)': total_cost
    })

df_current = pd.DataFrame(results).set_index('Cycle')

# --- Display results Current Policy---
print("\n=== Current Policy Simulation Results ===")
print(df_current)
print(f"\nAverage Total Cost (Current Policy): ${df_current['Total Cost ($)'].mean():.2f}")

```

Figure 5*Total replacement cost under Current Policy.*

=== Current Policy Simulation Results ===										
Cycle	Policy	Time of Failure	Fans Replaced	Arrival (min)	Repair Time (min)	Downtime (min)	Fan Cost (\$)	Tech Cost (\$)	Downtime Cost (\$)	Total Cost (\$)
1	Current	1200	2	20	30	50	64	25.0	500	589.0
2	Current	1700	1	20	20	40	32	20.0	400	452.0
3	Current	2200	1	20	20	40	32	20.0	400	452.0
4	Current	2500	1	20	20	40	32	20.0	400	452.0
5	Current	2900	1	20	20	40	32	20.0	400	452.0
6	Current	3400	1	20	20	40	32	20.0	400	452.0
7	Current	3500	1	20	20	40	32	20.0	400	452.0
8	Current	4500	2	20	30	50	64	25.0	500	589.0
9	Current	5000	1	20	20	40	32	20.0	400	452.0
10	Current	5500	1	30	20	50	32	25.0	500	557.0
11	Current	5900	1	20	20	40	32	20.0	400	452.0
12	Current	6300	1	30	20	50	32	25.0	500	557.0
13	Current	6900	1	20	20	40	32	20.0	400	452.0
14	Current	7500	1	20	20	40	32	20.0	400	452.0
15	Current	7800	1	30	20	50	32	25.0	500	557.0
16	Current	8100	1	20	20	40	32	20.0	400	452.0
17	Current	8500	1	20	20	40	32	20.0	400	452.0
18	Current	9500	1	20	20	40	32	20.0	400	452.0
19	Current	9600	1	30	20	50	32	25.0	500	557.0
20	Current	9700	1	45	20	65	32	32.5	650	714.5
21	Current	10700	1	20	20	40	32	20.0	400	452.0
...										
44	Current	21700	1	20	20	40	32	20.0	400	452.0
45	Current	22200	1	20	20	40	32	20.0	400	452.0

Average Total Cost (Current Policy): \$509.22

Proposed Policy: Replace all three fans whenever one fails.

```

# --- Proposed Policy Simulation ---

# Start with 3 new fans assigned random lifetimes
fan_lives = np.random.choice(lifetimes, size=3, p=lifetime_probs)
current_time = 0
results = []

# Run the simulation for the same numbers of cycles
for cycle in range(1, num_cycles + 1):
    time_to_failure = min(fan_lives)
    current_time += time_to_failure

    # Regardless of which fan failed, we replace all 3 fans every time.
    fan_lives = np.random.choice(lifetimes, size=3, p=lifetime_probs)
    fans_replaced = 3

    # fixed repair time and a random technician arrival time
    arrival_time = np.random.choice(arrival_times, p=arrival_probs)
    downtime = arrival_time + repair_time_fixed

```

```

fan_cost = fans_replaced * fan_cost_per_unit
tech_cost = (downtime / 60) * tech_cost_per_hour
downtime_cost = downtime * downtime_cost_per_min
total_cost = fan_cost + tech_cost + downtime_cost

results.append({
    'Cycle': cycle,
    'Policy': 'Proposed',
    'Time of Failure': current_time,
    'Fans Replaced': fans_replaced,
    'Arrival (min)': arrival_time,
    'Repair Time (min)': repair_time_fixed,
    'Downtime (min)': downtime,
    'Fan Cost ($)': fan_cost,
    'Tech Cost ($)': tech_cost,
    'Downtime Cost ($)': downtime_cost,
    'Total Cost ($)': total_cost
})

df_proposed = pd.DataFrame(results).set_index('Cycle')

```

Figure 6

Total replacement cost under Proposed Policy.

==== Proposed Policy Simulation Results ====										
Cycle	Policy	Time of Failure	Fans Replaced	Arrival (min)	Repair Time (min)	Downtime (min)	Fan Cost (\$)	Tech Cost (\$)	Downtime Cost (\$)	Total Cost (\$)
1	Proposed	1100	3	30	40	70	96	35.0	700	831.0
2	Proposed	2500	3	30	40	70	96	35.0	700	831.0
3	Proposed	3600	3	20	40	60	96	30.0	600	726.0
4	Proposed	4800	3	30	40	70	96	35.0	700	831.0
5	Proposed	6000	3	20	40	60	96	30.0	600	726.0
6	Proposed	7200	3	20	40	60	96	30.0	600	726.0
7	Proposed	8500	3	45	40	85	96	42.5	850	988.5
8	Proposed	9500	3	45	40	85	96	42.5	850	988.5
9	Proposed	10500	3	20	40	60	96	30.0	600	726.0
10	Proposed	11700	3	30	40	70	96	35.0	700	831.0
11	Proposed	12900	3	20	40	60	96	30.0	600	726.0
12	Proposed	13900	3	20	40	60	96	30.0	600	726.0
13	Proposed	15100	3	30	40	70	96	35.0	700	831.0
14	Proposed	16200	3	20	40	60	96	30.0	600	726.0
15	Proposed	17200	3	20	40	60	96	30.0	600	726.0
16	Proposed	18200	3	30	40	70	96	35.0	700	831.0
17	Proposed	19400	3	20	40	60	96	30.0	600	726.0
18	Proposed	20900	3	20	40	60	96	30.0	600	726.0
19	Proposed	21900	3	20	40	60	96	30.0	600	726.0
20	Proposed	23000	3	20	40	60	96	30.0	600	726.0
21	Proposed	24000	3	45	40	85	96	42.5	850	988.5
...										
44	Proposed	51000	3	20	40	60	96	30.0	600	726.0
45	Proposed	52200	3	30	40	70	96	35.0	700	831.0

Average Total Cost (Proposed Policy): \$803.00

C. Recommendation to Management

Based on the results, provides by Python script simulations, the recommendation is continuing with the current policy, where only the failed fans are replaced. Some reasons are as follows:

1. Lower average total cost:

- The Current Policy, which replaces only the failed fan, leads to significantly lower average total cost (about \$294 less per failure) compared to the Proposed Policy.

2. Higher economic efficiency:

- While the proposed policy aims to prevent future failures, the simulation indicates it leads to unnecessary replacement costs by discarding functional fans with remaining useful life.

3. Downtime and repair time already accounted for:

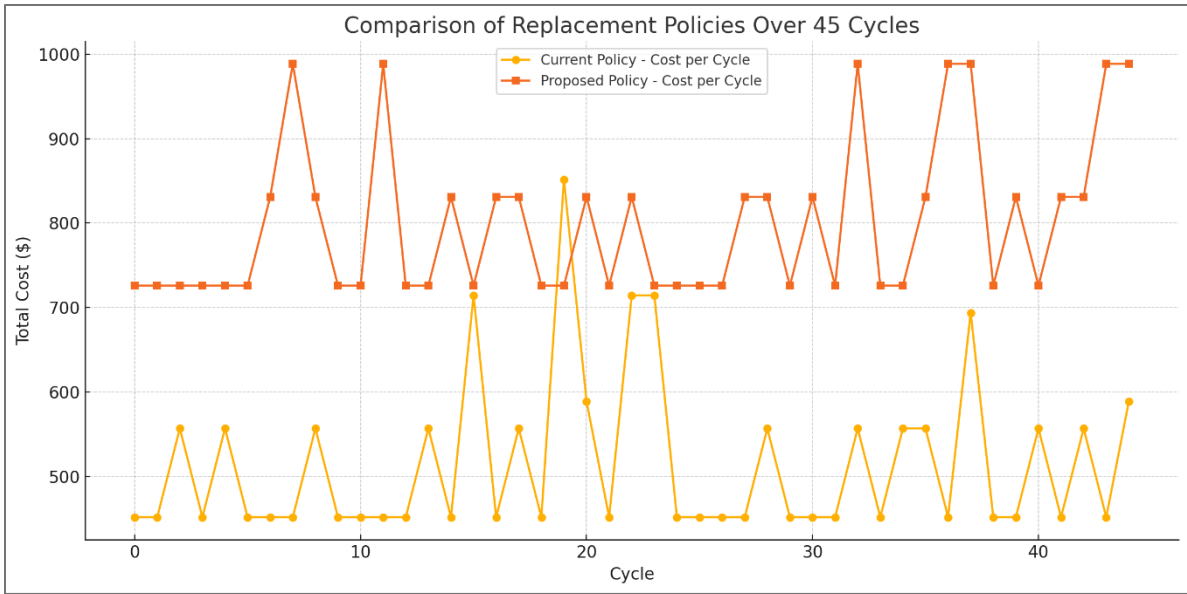
- Even after factoring in technician arrival delays and server downtime costs, the current policy remains more cost-effective.

4. Operational simplicity without financial penalty:

- The current approach avoids excessive interventions and optimizes resource use, without compromising service continuity, as evidenced by the simulation results.

Figure 7

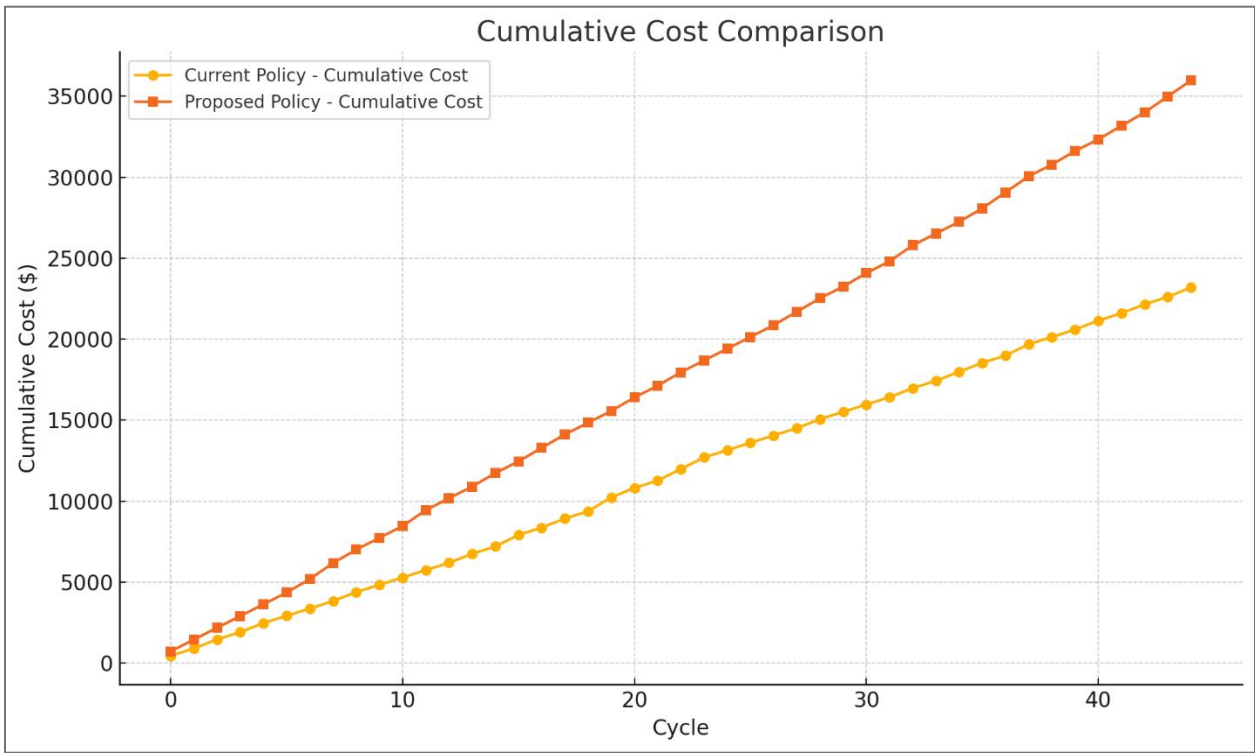
Comparison of Policies.



Note. The graph above shows us the comparison of the two policies and helps us graphically identify that replacing only 1 fan will ultimately give us a lower cost.

Figure 8

Cumulative Cost Comparison



Note. The line chart shows the cumulative costs between both proposals and as it can be seen the proposed policy increases with every trial given that each change means more money spent on the repairs as 3 fans will be changed every time which does not happen for the current policy.